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Improvement in Geothermal Well Integrity: A Review of New Advanced Cementing Technologies

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Abstract

Geothermal wells, with downhole conditions surpassing 350°C temperature, over 200 MPa pressure, and an extremely aggressive chemical environment, raise demands on conventional cementing systems for durability. This paper reviews the advanced cementing technologies developed to meet these challenges: improved thermal resistance, mechanical flexibility, and chemical stability. There is a discussion on novel formulation containing nano-silica, fibers, superplasticizers, and self-healing additives, improving thermal degradation resistance, flexing, and resistance against chemical attacks. New material geometries such as geopolymers, calcium aluminate cement, and chemical expanding agents have been identified that would provide superior performance for wellbore zonal isolation and long-term integrity of the well.

Operational efficiency will be further enhanced with well reliability through the integration of digital technologies, including AI-driven slurry optimization and real-time monitoring. The abovementioned developments are considered effective in reducing microcracking, debonding, and fluid migration to extend the lifespan of the geothermal wells by confirmation through field applications and studies in the laboratory. Finally, the environmental benefits accrue as reduced CO₂ emission and minimal groundwater contamination to the global goals of sustainability.

This review highlights the key deficiencies within the existing practices and presented guidelines for future research concerning materials development and advanced placement techniques. These advances would now allow access to more reliable, long-lasting, environmentally friendly operations of geothermal wells, which will enhance the adoption of geothermal energy as a clean resource.

Introduction

Cementing is the process of pumping a cement slurry into the wellbore to fill the annulus to the desired depth. Their main objectives in oil and gas wells are to safeguard and support the casing within the annulus, to prevent fluid migration through the space outside the casing, to prevent fluid flow into fractured formation and to seal off any abandoned sections of the well (Bourgoyne et al. 1986; H.J et al. 2020; Okoli 2018).

Geothermal energy is increasingly recognized as a critical component of the global transition to sustainable energy systems. Its high reliability and minimal environmental impact make it an attractive alternative to fossil fuels (Anderson and Rezaie 2019; Rybach 2003; Soltani et al. 2021). However, the long-term integrity of geothermal wells remains a significant challenge due to extreme conditions, including high temperatures, pressures, and chemically aggressive environments (Phi et al. 2019; Vrålstad et al. 2021). Effective well cementing is essential to maintaining wellbore stability, preventing fluid migration, and ensuring operational safety throughout the well's lifecycle (Al Ramadan et al. 2019).

Oil, gas, and geothermal wells experience a broad spectrum of temperatures, that ranges from below freezing to as high as 150°C (350°C -extreme case), and pressures varying from near ambient levels to more than 200 MPa (D. G. Bebout, R. G. Loucks, A. R. G 1979). Given the variable, complex, and challenging downhole conditions, pure cement alone cannot maintain the necessary integrity of the cement sheath along with the service life. As a result, different cement additives are used to modify the characteristics of the cement slurry. This ensures proper slurry placement, quick development of compressive strength, also effective zonal isolation and well integrity throughout the operational life of well (Agofack et al. 2019; Geoffrey et al. 2024; X. Wang et al. 2020; Y. Zhang et al. 2019).

If the formation is much stiffer than the cement, the chance of debonding between casing-cement and cement-formation would increase with the decrease in the temperature (Zheng et al. 2017). With the lower value of the thermal expansion coefficient of cement, if the cement is harder compared to the formation, there will be a possibility for debonding at the cement casing interface. However, in a depleted setting with declining pore pressure, if the formation is more rigid than the cement, the bonding between the cement and the formation as well as the casing and cement will break (Feneuil et al. 2023; Gheibi, Agofack, and Sangesland 2021; Lian et al. 2019; Y. Wu, Patel, and Salehi 2020).

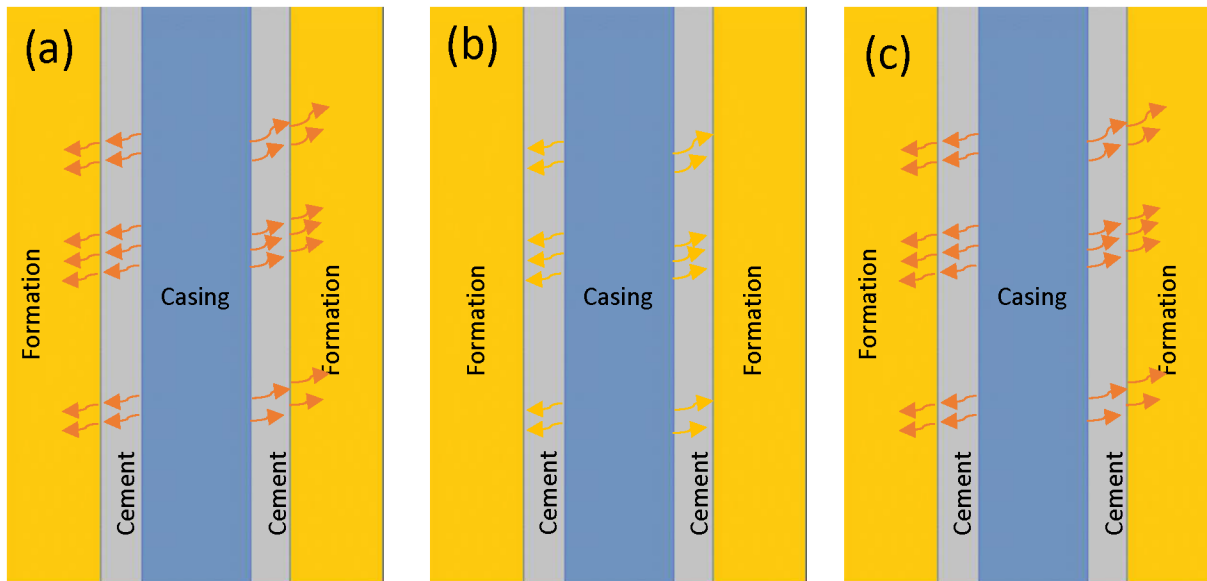


Figure 1—Deboding mechanisms at the casing–cement–formation interfaces under different geomechanical and thermal conditions.

- When the formation is significantly stiffer than the cement, a decrease in temperature increases the likelihood of debonding at both interfaces due to differential contraction.**
- Deboding in hard cement with a lower thermal expansion coefficient and higher stiffness than the formation leads to separation at the casing–cement interface**
- In a depleted reservoir with declining pore pressure, debonding occurs in rigid formations where both the cement–formation and casing–cement bonds are compromised.**

Therefore, matching the mechanical properties of cement and formation and that of the thermal expansion coefficient can avoid debonding risk in the casing cement and cement formation interfaces. A flexible and thermally adaptable type of cement applied with the optimized stiffness may be better fit for the characteristics of the formations with stable pore pressure operational conditions, thereby reducing a potential for bond failure.([Dou et al. 2020](#); [Y. Yan et al. 2021](#))

New improvements in cementing technologies target the performance of well cement under severe geothermal environments. New additions include thermally resistant additives, self-healing cements, and geopolymers for enhanced durability and flexibility in these applications. Such cementing technologies have been proven to counter specific challenges unique to geothermal environments by better resisting thermal stresses and chemical attacks ([Denduluri et al. 2023](#); [Shen et al. 2022](#); [S. Wang, Jian, et al. 2020a](#)).

Yet another promising field of innovation deals with nanomaterial inclusion in cement-based composites. Such nanomaterials, such as silica nanoparticles and carbon nanotubes, have proven quite promising to improve properties in cement materials, strength, permeability, and thermal stability, and these capabilities can indeed improve the bonding strength between the layers with a decreased micro-crack formation mechanism, hence a good contribution to keeping well integrity ([Luswata, Onyancha, and Thuo 2023a](#); [Sheng, Liyu, et al. 2020](#)).

Advances in techniques for placing cements improve geothermal well integrity. Techniques like MPC and real-time monitoring during cementing show promise for optimization of the placement of cements in a way to reduce damage to the formations. In this manner, the integrity of the cement sheath is more enhanced, and risks of failure minimized in more demanding geothermal conditions ([Rachimillah et al. 2021a](#); [Shryock 1984](#)).

This review discusses the latest development in cementing technologies meant to enhance the integrity of geothermal wells. Through novel materials, innovative techniques, and their field applications, the paper aims to give an overview of the current research state. The paper identifies gaps in knowledge and avenues for future development in dealing with the persistent challenges facing geothermal well cementing.

Challenges in Cementing Geothermal Wells

Extreme temperature, which can be very corrosive and destructive, severely affects the durability of the cement used in geothermal wells. The ability to understand how high temperatures vary in changing the mechanical and physical properties of cements is a prime importance factor in the development of assurance for the long-term reliability and safety of such structures ([Hager 2013](#); [Omotayo et al. 2016](#)). High temperatures will affect the microstructure of cement, particularly by increasing its porosity and permeability, which contributes to even weakening the mechanical integrity of the material ([Arioz 2007](#); [Qianmin et al. 2015](#)). Thermal strains and stresses develop within cement structures due to high temperature exposure and can cause cracking and spalling under proper conditions, especially in the case of high heating rates. At the cement-casing interface within wellbores, increases in temperature will strengthen the stresses and impact the comprehensive mechanical performance and integrity of the cement ([Janotka and Nürbergerová 2005](#); [Meng, Frash, and Carey 2023](#); [Omosebi et al. 2015](#)).

In geothermal wells, the alteration of the cement microstructure by chemicals from chloride, carbon dioxide, etc. undermines the durability of the cement. High chloride ion concentrations, such as are often found in brine fluids, can penetrate the cement sheath. ([Glasser, Marchand, and Samson 2008](#)) demonstrated that chlorides react with calcium hydroxide and other cement hydration products, which causes a reduction in alkalinity. As a result, the passive layer that protects the steel casing from corrosion is destabilized, which will further contribute to operational problems. In geothermal environments, particularly in CO₂ abundant reservoirs, carbon dioxide reacts with calcium hydroxide to form calcium carbonate via a reaction known as carbonation. While this might be interpreted as being favorable due to reduced permeability, over time, it will consume all binding phases like calcium silicate hydrates, C-S-H, from the cement, thereby reducing

its structural integrity. This is particularly catastrophic in geothermal wells where elevated temperatures enhance the rates of this reaction.(Papadakis, Vayenas, and Fardis 1991).

These microstructural changes, influenced by local ionic concentrations and pH, serve as evidence of the interaction complexity between the composition of cement and carbonation when it comes to geothermal conditions. The results indicate further studies and the need to tailor cement formulations, the role of various pozzolans, and real-time analysis of CO₂ exposure under high-pressure, high-temperature conditions in pursuit of enhanced cement performance for geothermal wells (Bjørge et al. 2019).

Carbonation has been reported to be a major issue with the geothermal well cementing process, especially for environments rich with CO₂. Works by Bjørge et al. 2019 showed that Portland cement with silica, while widely used to improve strength at high temperatures, results in carbonation much quicker compared to ordinary Portland cement (OPC). This enhanced carbonation can be due to the higher permeability and curing conditions, where a fully carbonated silica-cement sample was created under the same conditions in which OPC showed only a thin carbonation rim. The carbonation in cement progresses through successive steps: the dissolution of portlandite, the precipitation of calcium carbonate, and the formation of bicarbonate. Silica-cement tends to develop irregular transition zones with wormhole features characterized by stratified carbonate layers in aragonite-aragonite/vaterite.

The cement sheath must resist pressure-temperature cycles and corrosive reservoir fluids; hence, temperature prediction is of utmost importance in the process of cement design. Geothermal wells often have low fracture gradients and extensive lost circulation; hence, appropriate cementing designs and practices are always robust as a means of ensuring well integrity over a long period (Khizar et al. 2022).

The challenge of ensuring cement durability in geothermal wells, characterized by extreme temperatures and chemically aggressive fluids, has long raised questions about the adequacy of conventional oil well cements. Gallus, Watters, and Pyle answered this uncertainty with one of the first studies investigating the performance of different cements under realistic geothermal conditions. They subjected duplicate cement samples to geothermal fluids and temperatures in the Baca, NM, and Imperial Valley, CA, geothermal fields using a new testing method for as long as one year. They found that while conventional cements normally fail, it is possible to formulate cements that can survive the extreme conditions of geothermal wells for an estimated service life of 20 to 30 years. This research underlined the need for appropriate cement designs and testing protocols in furthering the reliability of geothermal well operations (Gallus, Watters, and Pyle 1979).

Ostroot and Shryock presented the challenges of cementing geothermal wells with ultra-extreme bottom-hole static temperatures of 500°F to 700°F such as those encountered in California's Salton Sea area. Their paper placed great emphasis on thorough planning, cooperation between the producers, drilling contractors, and service companies, and the importance of laboratory testing to simulate well conditions. They applied appropriate cement formulations through cementing programs in eight geothermal steam wells and demonstrated a preference for API Class G over Class E, with the addition of strength-stabilizing additives such as silica flour and expanded perlite for thermal insulation. This approach ensured adequate thickening time, long-term compressive strength, and minimal permeability, even under extreme thermal stresses. Besides, the study outlined that full cement returns are quite critical in casing creep reduction to avoid such incidents of sudden failure. The work considerably furthered high-temperature geothermal well cementing (Ostroot 1964).

Geothermal reservoirs, with their high temperatures and aggressive, highly corrosive conditions, present extreme challenges for cement, substantially shortening the durability of cements and limiting the applicability of conventional oil-field cement formulations. Geothermal wells can be incredibly hot, with temperatures of up to 300°C for conventional wells. Lately, ultra-high-temperature or supercritical geothermal reservoirs, which have a temperature over 500°C, have been in development. Recently, supercritical systems have garnered considerable attention because they enable enhanced heat recovery

compared with conventional geothermal reservoirs, hence offering a more effective and sustainable energy source (Mortensen et al. 2014).

Geothermal cements are under constant temperature stresses due to the temperature fluctuations of the well. For instance, when cold water is injected, casing contracts because of cooling and develops tensile stresses that can be responsible for the creation of micro-annuli at the cement sheath casing interface. During hot water recovery, on the contrary, there is thermal dilatation whereby compressive stresses on the cement sheath may be responsible for the creation of microcracks. These repeated temperature variations highlight the need for robust cement designs that are able to resist such thermal cycling and maintain well integrity (Elders, Friðleifsson, and Pálsson 2014).

In geothermal wells, a defective cement sheath can have very significant consequences, as the loss of well integrity fails in providing protection to the casing or maintaining zonal isolation. In general, the detection and repair of any failures in cement within underground environments is difficult; therefore, self-healing cement and self-re-adhering cement are very desired for geothermal reservoirs. Most cement types display an inherent self-healing tendency, especially in their first days after hardening. These properties are due to a variety of factors, such as continued slow hydration reactions which strengthen the material over time, unreacted particles in the cement matrix that become activated upon exposure to fluids, and interactions between hydrated products and environmental constituents to form crack-sealing phases that lead to partial strength recovery. In fact, such self-healing mechanisms provide promising ways for sustaining long-term integrity of wells in geothermal systems (Huang, Ye, and Damidot 2013; Teodoriu and Falcone 2009).

The ability of cements of self-healing is, as a matter of fact, known as an autogenous healing. In order to have more pronounced self-healing, slow-reacting components are intentionally introduced into cement formulation; this method is termed autonomic healing. Pozzolans-secondary cementitious materials-naturally occurring and industrial by-products-assist in self-healing through pozzolanic reactions in alkaline Portland cement slurries or lime/alkali activator. Cement-based materials can undergo early age cracking due to shrinkage under restrained conditions. These include clays,; fly ash, types F or C, FAF, and FAC, respectively,; granulated blast furnace slag, and amorphous silica (Ahn and Kishi 2010; Huang, Ye, and Damidot 2013; Qian, Zhou, and Schlangen 2010; Sahmaran, Yildirim, and Erdem 2013; Termkhajornkit et al. 2009).

Innovative Cementing Techniques

Advanced Cement Formulations

The harsh conditions of high temperature and high pressure (HTHP) in oil wells often cause significant strength retrogression and structural degradation in the cement sheath, compromising well integrity. To address this, silica flour is widely used as a thermal stabilizer in cement formulations. A comprehensive study by Jiang et al. 2021 explored the impact of silica flour with different particle sizes on the long-term thermal stability of cement pastes, providing critical insights into optimizing cement performance under extreme conditions(Jiang et al. 2021). Under HTHP cementing conditions, structural deteriorations in the cement sheath and loss of cement strength can occur for oil wells. Silica flour is normally added to increase thermal stability in cement. Among other normal pastes, cement pastes with 200-mesh silica flour have exhibited the highest compressive strengths after HTHP curing (Jiang et al. 2021; Krakowiak et al. 2018).

Among other classes of cements, Class G cement is the most commonly used in oil well cementing, and under the condition of high temperature and CO₂, its degradation significantly increases the occurrence of accidents. On the contrary, Calcium aluminate cements shows good resistance in both conditions of high temperature and corrosion (G. Wu et al. 2022). The study investigates by (Khan et al. 2023) the behavior of calcium aluminate cement as a substitution to OPC in ultra-high-performance concrete at exposed elevated temperatures to manufacture a non-explosive spalling high-performance concrete. It was observed that CAC-based ultra-high-performance concrete and OPC-based ultra-high-performance concrete attain their

maximum strength at the same temperature of 250 °C, though the CAC-based ultra-high-performance concrete showed the peak strength as high as 153.8 MPa, much higher than the peak strength of the OPC-based ultra-high-performance concrete, which was 137.5 MPa(Khan et al. 2023)

A reduced water-to-cement ratio helps increase the strength and density of the cement used. To the contrary, special additives like superplasticizers maintain its work ability while keeping the water content low (Jhatial et al. 2018; Smirnova 2016). It can contain expansive agents, such as calcium sulfoaluminate or magnesium oxide, that counterbalance the shrinkage that may occur during curing at high temperature. Under hydration, these expansive agents produce controlled expansion, thus eliminating potential microcracking and improving long-term integrity (Guo et al. 2021; Mehdipour and Khayat 2018; Wyrzykowski, Terrasi, and Lura 2018). Some cement systems take advantage of thermal activation, whereby the development of strength is accelerated by elevated temperature, using additives that increase the rate of reaction or phases formed which are thermally stable (Razaul, Huzaifa, and Hashim 2016; C. L. Yong, Mo, and Koting 2023).

Alternative Cement Blends.

Pozzolanic Cements:

Recently, pozzolans such as fly ash or volcanic ash have been in the focus as an alternative to cement for thermal stability. These materials contribute to the improvement of thermal stability and, at the same time, reduce the amount of CO₂ emitted, thus representing environmental sustainability (Mousavinezhad et al. 2023).

The slower crystallization of ettringite may be due to two reasons, as compared to that in cement paste. First, very high pozzolan activity is there in fly ash, due to which the consumption of calcium hydroxide in the reaction forms C-S-H gel (Terzić et al. 2015). Not only the available calcium hydroxide is consumed by this reaction but the unhydrated mineral particles also get covered that prolongs the "hydration induction period." Therefore, the initial stages of crystallization of ettringite were retarded, that impinged on its crystallization.

Nano-silica:

Owing to the high surface energy and pozzolan effect, nanomaterials can rapidly react with cement hydration products like calcium hydroxide, especially nano silica, to form C-S-H gel and other products that increase the cement slurry strength.

According to the studies by (S. Wang et al. 2020), additions of nano silica in the range of 0-3% result in an apparent improvement in compressive strength of the cement paste. Addition of nano silica up to the specified optimum content level effectively improves the compressive strength of the cement stone (Riahi and Nazari 2012). For example, in the case of activated slag composites, the compressive strength at 72 hours increases from 35.8 MPa to 40.9 MPa upon the addition of 2% nano silica, which is equivalent to an increase of 14.2%, as reported by (Behfarnia and Rostami 2017; Long et al. 2018; S. Wang et al. 2020).

Geopolymer cements:

Geopolymer cements have recently gained a lot of attention as promising alternatives to Portland cement in the most challenging environmental conditions, including those related to geothermal well cementing. Geopolymers are considered a new generation of cement, taking into consideration increased thermal stability and chemical resistance, specifically in extreme HT-HP conditions. In the study by Horan et al. (2022), geopolymers were found to have outstanding potential for application in geothermal fields because of their ability to handle extreme geothermal environments where conventional cement materials normally lose their bonding capability (Horan et al. 2022). Since these geopolymers retain most of their structure at more than 300°C, they have typically been preferred for geothermal wells that undergo very high thermal stress alongside an aggressive chemical environment.

There are publications proving the success of geopolymers in geothermal reservoirs. Many experiments were done to investigate how fly ash-based geopolymers behave in simulated geothermal situations. Many

research papers have proven geopolymers can be effective in geothermal wells. Results show from (Sarker, Kelly, and Yao 2014) that geopolymer from fly ash withstood 1000°C temperature and had just 4.8% mass loss, compared to the major mass loss (90%) of ordinary Portland cement (OPC). (Dombrowski, Buchwald, and Weil 2007) made a geopolymer using fly ash and explored how higher amounts of calcium improved its performance in high temperatures. When geopolymer was heated past 600°C, it turned into crystallized sodalite and amorphous aluminosilicates which then became nepheline and caused the geopolymer to arrange more closely over time. Nepheline's temperature rise changed it into albite. Higher strength and smaller shrinkage were seen in 8% Ca(OH)₂ geopolymer when nepheline was heated to approximately 800°C and feldspar to 1000°C.

The results showed that geopolymers had excellent resistance to acidic geothermal fluids that have so far corroded conventional cements, while they maintained their strength through a protracted exposure time at temperatures greater than 200°C. According to the authors, geopolymer cements could be used in a big way to enhance the durability and long-term reliability of geothermal well casings (Zhuang et al. 2016).

During the early phase, investigators found that class F fly ash geopolymer cement might be suitable for cement production. It is apparent from the materials studied that these substances do not let oilfield areas become contaminated, stand up to being exposed to acids, fix split parts in wells and are more tolerant to higher temperature and pressure than other types of cement. In addition, materials made with silicate-activated geopolymers can withstand temperatures up to 800°C and find use in oil and gas as well as geothermal energy wells. In particular, Geopolymer binders (Duxson et al. 2007) save more than 80% of the CO₂ emissions that are released by making OPC

Resistance to Sulfate and CO₂ Attack. The performance characteristics enhanced by supplementary cementitious materials addition, such as silica fume and blast furnace slag, include increased resistance to sulfate attack and exposure to CO₂. These properties are particularly important in geothermal wells and CO₂ injection wells, where these previously mentioned effects can substantially degrade conventional Class G cement (Maria C.G.J, Ruben, and Susan A.B 2019).

Ground Granulated Blast-Furnace Slag (GGBS) is one of the best cementitious materials and finds major applications in concrete. GGBS-Slag is obtained as a by-product of reduction of iron ore to iron in a blast furnace. The temperature of blast furnace rises as high as 1500°C. Iron ore is mixed with coke and limestone and then fed into the furnace; the iron ore is thereby reduced to iron. The other materials in the substance combine to form a slag that floats to the top. This molten slag is removed and cooled rapidly by being plunged into vast volumes of water, which is called quenching. This rapid cooling increases the cementitious properties of the material, creating particles that are similar in appearance to coarse sand. Once cooled, the slag is then dried and ground into a fine powder, which is the GGBS. GGBS, often otherwise referred to as slag cement, has many of the characteristics of cement clinker and proves to be a good replacement or addition in concrete mixes. Again, this not only avoids waste from the iron industry but also provides a kind of greener concrete at lower environmental cost than traditional manufacturing of cement (Arif et al. 2020; Kansal, Singla, and Garg 2020).

Silica fume is a byproduct from the manufacturing of silicon metal or ferrosilicon alloys. One of the most valuable uses of silica fume is as an additive in concrete because of its biochemical and physical characteristics make it an extremely reactive pozzolan. Concrete made with silica fume develops very high strength. On the other hand, silica-fume concretes place additional demands on mixing, placing, and curing by the concrete contractor. Silicon metal and alloys are produced in electric furnaces using quartz, coal, and wood chips as the major raw materials. Silica fume is an inorganic additive in concrete and is basically composed of non-crystalline silicon dioxide. Because of its fineness, large surface area, and high silica content, it acts as a reactive pozzolan, improving the performance of the concrete (Wani and Bhandari 2021; Zięba, Witek, and Mońka 2018).

Self-Healing Cement. The self-healing cements have the capability of automatically repairing microcracks, since they are incorporated with materials that are to react with environmental elements to form new sealing products. This innovation is targeted at offering better well integrity in the long term, particularly under the conditions of HTHP where microcracking can be more pronounced (Mario et al. 2013).

Recently, researchers focused on new materials and methods to enhance structural sustainability. Among the recent developments are self-healing technologies, which probably represent one of the most promising alternatives for extending concrete structure lifetimes. The first practical observation of self-healing within concrete dates back more than twenty years. Evidence supporting this technology shows that it largely enhances durability with minimal use of maintenance. It has been identified from studies that concrete cracks naturally heal when moisture interacts with un-hydrated cement particles to form new bonds. Conversely, modern construction practices have reduced the natural self-healing capacity of concrete, such as by reducing cement content.

The concrete-based composite self-healing may be either autogenous or autonomous in character. In an autogenous approach, the un-hydrated cement clinker within the mix becomes the healing agent. In contrast, the autonomous approach intends to incorporate one or more synthetic additives or microorganisms, such as calcite-precipitating bacteria, into the binder mix that activates upon rupture of capsules or ingress of moisture/oxygen into the structure, thereby repairing the cracks (Nair, Gupta, and Agrawal 2022).

The research work investigated by (Qureshi, Kanellopoulos, and Al-Tabbaa 2018) showed the impact of expansive minerals, like magnesium oxide, bentonite clay, and quicklime, on the early-age autogenous self-healing capability of Portland cement paste. The effects of single and combined multiple mineral dosages in PC was analyzed in detail, together with a brief state-of-the-art review concerning autogenous self-healing and the application of minerals. The dominant constituents of healing materials are hydrated/swelling products of expansive minerals. The application of mineral-based mixes succeeded in sealing a 180- μm crack width in 28 days. Crack bridging in other words restoration of strength, crack sealing, which is physical closure by crystallization besides improved durability performance drove this self-healing process (Qureshi, Kanellopoulos, and Al-Tabbaa 2018)

Nanomaterials for Improved Durability. Addition of nanoparticles to the cement formulation, such as nano-silica and carbon nanotubes, has indicated positive responses towards compressive strength, thermal stability, resistance to crack, and modification of the microstructure of wellbore cement. It leads to a more homogeneous hydration of cement with reduced permeability (Yonathan 2018). Since the emergence of nanotechnology in construction, many kinds of nanomaterials have been added to concrete applications. As discussed previously, nanomaterials have great potential to improve the performance and durability of concrete significantly. This section will further discuss the types of nanomaterials explored by past researchers and their applications in Ultra-High Performance Concrete (UHPC).

Silica and alumina are both very important in the hydration of cement for different purposes. Silica transmits strength properties, whereas alumina primarily influences the time taken for setting. Nano alumina is more developed than regular alumina; though promising in concretes, its application in UHPC is relatively under-explored. Nano alumina UHPC reduces the initial set drastically. The rapid setting inhibits segregation and flocculation, and the mix turns out to be uniform without disturbances that might degrade the performance of UHPC (Hosseini et al. 2014; Nazari and Riahi 2011).

Nano-alumina is used as a dispersion agent to improve the uniformity of cement particle dispersion. In addition, due to the nanoscale size of nano-alumina, it is possible to occupy the holes within the hydration gel. In UHPC, in which the content of cement is high, an efficient dispersion of cement particles and silica activity during hydration are particularly significant. Without the refining influence of nano-alumina, hydration proceeds more slowly since silica cannot easily penetrate the dense gel structure. Nano-alumina provides pathways in the gel through which silica and other binders can penetrate to strengthen it (Rawdhan and Al-Jelawy 2020).

Apart from accelerating hydration, nano-alumina optimizes the packing density of UHPC particles, reducing porosity and boosting durability. Its role in refining the hydration gel makes concrete denser and more resilient. For this reason, it represents an essential innovation in that it could be critical to meeting high-performance requirements in UHPC applications. It performs a double function: setting acceleration, as well as structural enhancement. For this reason, nano-alumina has become a key ingredient in modern cement formulations (Jo et al. 2007)

Most importantly, Carbon Nano-Tubes (CNT) usage in UHPC provides many benefits regarding flexibility. CNT is capable of adaptive design that may allow UHPC for either unique or rigid applications. Such flexibility adds to strength and makes CNT one of the most functional nanomaterials in improving both flexibility and strength of UHPC compared to other alternatives. Besides, CNTs are much smaller in volume and size compared to most other nanomaterials, hence allowing for better integration (Rostamiyan et al. 2015)

The major effect of CNT in UHPC is the enhancement of tensile and compressive strength. In fact, CNTs can even replace traditional steel reinforcement, thus enabling UHPC to bear higher loads. This substitution will not only reduce heavy reinforcement but will also make lighter structures, possibly cutting construction times and costs. Despite these advantages, the limited availability of CNTs, their high cost, and the lack of expertise and standardized guidelines for their incorporation into UHPC remain challenges, limiting the widespread adoption of CNTs in UHPC mixes (Morsy, Alsayed, and Aqel 2011).

Lost Circulation Resistance. Addition of fibers made of fibers such as polypropylene or glass to the cement formulation is done with the purpose of improving resistance to lost circulation. Fiber-reinforced cements promise greater mechanical stability, reducing fractures and wellbore failures in geothermal and deep oil wells (Anas et al. 2018).

The study by Ying Ji, Qianqian Sha, Gang Zhu, Yuze Xue, and Tinghui Zhang examines the challenges experienced in significant heat loss as well as low heat recovery efficiency in shallow geothermal cement intended for heat exchange in geothermal wells. To overcome these problems, researchers added thermal insulation materials, straw ash, and natural zeolite as partial substitutes for cement. They also introduced new activators of alkalis, such as water glass. This reduced the value of thermal conductivity of cement, keeping all other mechanical characteristics constant. The work further attempted to scrutinize how the use of alkali activators changes the macroscopic properties such as compressive strength, thermal conductivity, fluidity, thickening time, and micrometrics such as porosity and total structural parameters (Ji et al. 2024).

Graphene oxide (GO) has been researched as the car of deep eutectic solvents (DES). Based on this design concept, Wenxi Zhu et al. prepared graphene oxide nanofluids using DES as the GO carriers. The GO's dispersed ability could be improved as the GO consumption and costs associated with it were simultaneously reduced. The innovative application of GONF in enhancing the thermal conductivity of geothermal cement was focused in this study. The authors used this method to solve the problems of lowered strength of cement and fast loss of fluidity with traditional high-thermal conductivity materials, thereby making the geothermal application more effective (Zhu et al. 2024).

A study by Marita L. Berndt and Aristodimos J. Philippacopoulos also indicates that round steel fibers of 13 mm length have proved to be significantly effective in enhancing properties at relatively low volume fractions of 0.5% and 1%. In contrast, steel microfibers required a higher volume fraction of 5% to achieve notable improvements in tensile strength. Carbon microfibers also demonstrated the ability to enhance tensile properties when used at volume fractions of 2% and 5%. However, glass and basalt fibers were found to be ineffective under the conditions tested. The addition of round steel fibers also improved compressive strength, while steel microfibers, particularly at a 5% volume fraction, increased both thermal conductivity and the coefficient of thermal expansion. Importantly, steel fibers did not negatively impact the coefficient of permeability, making them a viable option for improving material performance (Berndt and Philippacopoulos 2002)

High-Performance Cement Design. Advanced cement formulations intended for geothermal wells specifically focus on hydration kinetics for High-Temperature and High-Pressure (HTHP) conditions. Geothermal wells are exposed to extreme thermal and mechanical stresses; therefore, the cements used must resist these conditions in a manner that ensures well integrity over the long term. It has been found that common cements generally fail to provide stability, especially at the stage of hydration-the key factor in early failures or reduced performance over time. Advanced cements shall have formulations that allow for certain specific hydration kinetics under HTHP conditions. For this purpose, modifiers of the rate of cement setting and hardening at high temperatures shall be used with the aim of superior long-term performance and not only avoiding early time failures (Xuerui et al. 2020).

The paper of Ramaswamy Gautam, Abhinav Hazra, Prashant Faujdar, Suvendu Sen, Bibhash Chandra Mishra, Tushar Sharma, and Shailesh Kumar discusses the challenges in cement slurry design, where precise engineering practices and appropriate cementing operations are required for satisfactory performance. The study also uses simulation to determine sensitivity of retarders in choosing thickening time, optimising cement properties, and also predicting the behavior of cement slurries under HPHT conditions. The results showed that a modified Magnesium Oxide slurry, which is a blend of water-glass and magnesium oxide, was found to be the best candidate for HPHT slurry design and, hence, a denser cement paste matrix was achieved. A retarder was used to shorten the thickening time, whereas the addition of a water-glass solution improved the thickening time of Magnesium Oxide cement. These demonstrated that thickening time is a function of both exposure duration and temperature. Further, the study also found out that under HPHT conditions, the rheological properties of the slurry such as plastic viscosity, gel strength, and yield point were decreased. Finally, the paper identifies one of the optimal cement slurry design strategies for various downhole applications in HPHT wells (Gautam et al. 2024).

As listed below, Table 1 summarizes numerous advanced types of cement with their specific applications discussed in the above detailed discussion for the geothermal well environment. It draws out the exceptional properties and performance of every cement type such as stability in temperature, chemical attacks, and improvement in mechanical strength. The references given for every type point out an important study which underlines how useful these cement types might be. The table also indicates the temperature ranges, which are suitable for high-temperature, high-pressure (HTHP) geothermal conditions. This is a resource that can be used as a reference when choosing the best cementing solutions for geothermal energy production challenges.

Table 1—Advanced Cement Formulations for High-Temperature, High-Pressure (HTHP) Conditions

Cement Type	Description	References	Temperature Range
Silica Flour-Based Cement	Silica flour is used as a thermal stabilizer in cement formulations to improve long-term thermal stability under HTHP conditions. Cement pastes with 200-mesh silica flour show the highest compressive strengths after HTHP curing.	Jiang et al. (2021); Krakowiak et al. (2018); Tao et al. (2021)	Up to 250°C (based on silica flour use)
Calcium Aluminate Cement (CAC)	CAC-based cement shows excellent resistance to both high temperature and corrosion (particularly CO ₂ exposure), superior to Class G cement. CAC-based ultra-high performance concrete shows peak strength at 250°C.	Wu et al. (2022); Khan et al. (2023)	Up to 300°C (based on CAC studies)
Geopolymer Cement	Geopolymers, particularly those based on fly ash, exhibit exceptional thermal stability and chemical resistance in geothermal and HTHP environments, withstanding temperatures greater than 300°C. They are ideal for geothermal wells.	Horan et al. 2022, Sarker, Kelly, and Yao 2014, Dombrowski, Buchwald, and Weil 2007, Zhuang et al. 2016	300°C to 800°C

Pozzolanic Cements (Fly Ash/Volcanic Ash)	Pozzolans like fly ash and volcanic ash improve thermal stability and reduce CO ₂ emissions. They also enhance cement's compressive strength and longevity under HTHP conditions.	Riahi and Nazari 2012, S. Wang et al. 2020 ,Terzić et al. 2015	Up to 250°C (Variable, based on blend composition)
Nano-Silica Cement	Nano-silica improves the compressive strength of cement slurry, with optimum additions increasing strength by up to 14.2%. It accelerates hydration and enhances the microstructure.	S. Wang et al. 2020, Behfarnia and Rostami 2017; Long et al. 2018; S. Wang et al. 2020	up to 250°C (typical range)
Self-Healing Cement	Self-healing cements automatically repair microcracks by utilizing expansive minerals or microorganisms that react with environmental elements. This technology enhances long-term well integrity.	Qureshi et al. (2018); Mario et al. (2013)	Up to 300°C (Applicable in temperature ranges, especially under HTHP conditions for crack repair)
Nanomaterials for Durability	The addition of nanomaterials like nano-silica and carbon nanotubes enhances compressive strength, crack resistance, thermal stability, and microstructure. Nano-alumina and carbon nanotubes (CNT) are critical in improving the mechanical properties of UHPC.	Yonathan (2018); Hosseini et al. (2014); Morsy et al. (2011); Rawdhan and Al-Jelawy (2020)	Up to 250°C (depending on material)

Lost Circulation Resistance Cement	Fiber-reinforced cements, including those with polypropylene or glass fibers, are used to enhance resistance to lost circulation, improving wellbore stability. Thermal conductivity is improved with additives like straw ash or zeolite.	Anas et al. (2018); Ji et al. (2024); Zhu et al. (2024); Berndt and Philippacopoulos (2002)	Up to 150°C (for geothermal applications)
Magnesium Oxide Cement (HPHT Slurry)	Modified magnesium oxide cement slurry shows promising performance for HPHT conditions, with optimized thickening time and better slurry properties.	Gautam et al. (2024)	Up to 350°C (HPHT conditions)

Use of Polymers and Nanomaterials

Latex polymers like styrene-butadiene latex are put to very good use in improving elasticity, bonding, and flexibility of cement-cardinal for preventing microcracking and enhancing zonal isolation (Romildo and Jo 2018; Z. Yong et al. 2023). During setting and curing; Acrylic polymers are used to enhance water retention in cement slurries, reduction of shrinkage, as well as enhanced tensile strength along with resistance to chemical degradation (Haixin et al. 2020; H. Huajie et al. 2016). The superabsorbent polymers serve to absorb water from the cement slurry and swell; thus, they act against shrinkage. This assists a lot in high-temperature cementing, where shrinkage could result in gaps that develop into well integrity issues (L. Huajie et al. 2016; Zhong et al. 2019).

Nano-silica happens to be one of the most common nanomaterials that are being applied in cementing oil wells. It is meant for improvement in compactness as well as enhancement in the density of the cement matrix, which results in higher compressive strength along with improved resistance to chemical attack (Akshar et al. 2020; Ramana, Murthy, and Faruq 2020; Xiaoliang et al. 2023). Carbon nanotubes reinforce the cement matrix at a nanoscale level. It improves the mechanical properties of cement by giving out outstanding strength, including flexural and tensile strength (Laura and Philippe 2020; Simone et al. 2009; Vanessa et al. 2019). Graphene oxide, with excellent mechanical and thermal properties, has great potential to enhance the strength, durability, and thermal stability of cement significantly. It is generally used to improve the bonding strength of the cement matrix with casing (Li et al. 2020; Wenguang et al. 2018).

Table 2 offers an in-depth summary of the above discussion on advanced materials incorporated into cement, with relevance to geothermal applications. Materials include latex and acrylic polymers, superabsorbent polymers, nano-silica, carbon nanotubes, and graphene oxide, with descriptions and key benefits and operational ranges, how each material contributes to improved performance in cement under geothermal conditions, such as elasticity improvement, chemical resistance, thermal stability, and mechanical strength. Including the range of temperatures and pressures gives practical insight into the use of these materials in high-temperature, high-pressure geothermal settings.

Table 2—Polymers and Nanomaterials for Enhancing Cement Properties in Oil Well Cementing

Material Type	Description	References	Key Benefits	Temperature Range	Pressure Range
Latex Polymers (Styrene-Butadiene Latex)	Styrene-butadiene latex improves elasticity, bonding, and flexibility of cement, preventing microcracking and enhancing zonal isolation.	Romildo and Jo (2018); Yong et al. (2023)	Improves elasticity, bonding, and flexibility; prevents microcracking; enhances zonal isolation	Up to 180°C (typical for oil well cementing)	Up to 20,000 psi (typical pressure for cementing)
Acrylic Polymers	Acrylic polymers enhance water retention in cement slurries, reduce shrinkage, and improve tensile strength along with resistance to chemical degradation.	Haoxin et al. (2020); H. Huajie et al. (2016)	Enhances water retention; reduces shrinkage; improves tensile strength and resistance to chemical degradation	Up to 200°C (typical for geothermal applications)	Up to 15,000 psi (for geothermal conditions)
Superabsorbent Polymers	Superabsorbent polymers absorb water from the cement slurry and swell, counteracting shrinkage and preventing gaps that could lead to well integrity issues in high-temperature cementing.	L. Huajie et al. (2016); Peihua et al. (2019)	Prevents shrinkage; improves well integrity at high temperatures	Up to 250°C (for geothermal applications)	Up to 15,000 psi (for geothermal conditions)

Nano-Silica	Nano-silica enhances cement matrix compactness and density, improving compressive strength and chemical resistance.	Akshar et al. (2020); Ramana, Murthy, and Faruq (2020); Xiaoliang et al. (2023)	Improves compactness, density, compressive strength, and chemical resistance	Up to 250°C (for geothermal applications)	Up to 20,000 psi (for geothermal conditions)
Carbon Nanotubes	Carbon nanotubes reinforce the cement matrix at a nanoscale level, enhancing mechanical properties such as flexural and tensile strength.	Laura and Philippe (2020); Simone et al. (2009)	Enhances mechanical properties (flexural and tensile strength)	Up to 250°C (for geothermal applications)	Up to 20,000 psi (for geothermal conditions)
Graphene Oxide	Graphene oxide improves the strength, durability, and thermal stability of cement, and is used to enhance bonding strength between the cement matrix and casing.	Li et al. (2020); Wenguang et al. (2018)	Enhances strength, durability, thermal stability, and bonding with casing	Up to 250°C-270°C (for geothermal applications)	Up to 20,000 psi (for geothermal conditions)

Application of Expanding Cements

Chemical expanding agents are added in cement slurry along with some chemical expanding agents that will undergo some controlled expansions due to certain chemical reactions during hydration (Hao et al. 2019). Calcium sulfoaluminate expands by the formation of a crystalline compound called ettringite, which grows in volume during the hydration of CSA-based cement. The rate of formation of ettringite is rather fast, giving early expansion that offsets shrinkage (Frank and Barbara 2010; Solmoi et al. 2021; Tang et al. 2015).

Magnesium Oxide reacts with water to form a magnesium hydroxide, $\text{Mg}(\text{OH})_2$, which has a larger molar volume and causes expansion. Such reaction produces some sort of delayed expansion that compensates for long-term shrinkage of cement (Lei and Bing 2021; Ye et al. 2021). The aluminum powder reacts with the hydroxide ions, which are one of the products of cement hydration, producing hydrogen gas. This hydrogen forms bubbles within the cement matrix, creating a slight expansion (Shingo et al. 2013). The reaction of sodium sulfate with the calcium aluminates in the cement forms expansive ettringite. This reaction generates early expansion, much like CSA, which counteracts the shrinkage during hydration (Haixin et al. 2021; Jun et al. 2020).

Table 3 discusses cement expanding agents, which are meant to increase the durability and integrity of cement under conditions of geothermal well applications. It talks about materials such as calcium sulfoaluminate (CSA) and sodium sulfate that cause early expansion to counteract shrinkage during hydration. On the other hand, magnesium oxide is responsible for delayed expansion, offsetting long-term shrinkage, and it can function well under extreme conditions, like high-temperature and high-pressure environments. Aluminum powder causes controlled expansion through the production of hydrogen gas and improves the adaptability of the cement. The table also shows the major advantages and operating ranges of each agent, ensuring their ability in geothermal applications.

Table 3—Chemical Expanding Agents for Shrinkage Compensation in Cement Slurry

Cement expanding agents	Description	References	Key Benefits	Temperature Range	Pressure Range
Calcium Sulfoaluminate (CSA)	Calcium sulfoaluminate expands by forming ettringite, a crystalline compound that grows in volume during hydration, providing early expansion to counteract shrinkage.	Frank and Barbara (2010); Solmoi et al. (2021); S. W. Tang et al. (2015)	Provides early expansion, offsets shrinkage	Up to 250°C (for geothermal applications)	Up to 15,000 psi (for geothermal conditions)
Magnesium Oxide	Magnesium oxide reacts with water to form magnesium hydroxide ($Mg(OH)_2$), causing delayed expansion to compensate for long-term shrinkage.	Lei and Bing (2021); Yuxun et al. (2021)	Compensates for long-term shrinkage	Up to 350°C (for geothermal applications)	Up to 25,000 psi (for geothermal conditions)
Aluminum Powder	Aluminum powder reacts with hydroxide ions produced during cement hydration, generating hydrogen gas that forms bubbles, causing slight expansion within the cement matrix.	Shingo et al. (2013)	Provides slight expansion via hydrogen gas	Up to 200°C (for geothermal applications)	Up to 15,000 psi (for geothermal conditions)
Sodium Sulfate	Sodium sulfate reacts with calcium aluminates in cement to form ettringite, generating early expansion similar to CSA, which counteracts shrinkage during hydration.	Haoxin et al. (2021); Jun et al. (2020)	Provides early expansion, offsets shrinkage	Up to 250°C (for geothermal applications)	Up to 15,000 psi (for geothermal conditions)

Fiber-Reinforced Cements

The fibers improve the toughness, tensile strength, impact resistance, and fracture toughness of the cement, which helps in making the cement more resistant to crack, thermal cycling, and mechanical stresses during well operations ([Chunmei et al. 2020](#); [Mohtasham Moein et al. 2022a](#); [Shuai, Erol, and Weidong 2019](#)).

Depending on well conditions given and on required mechanical properties, different types of synthetic fibers are used in fiber-reinforced cements. Polypropylene fibers are lightweight, chemically resistant, and economical. They enhance the cement toughness and crack resistance without seriously affecting the weight and density of the cement ([Hua et al. 2018](#); [Jing jun et al. 2016](#)). Nylon fibers exhibit high tensile strength and flexibility, which enable them to enhance the toughness and crack resistance of cement. Besides that, they improve impact resistance and provide greater ductility to the cement in general ([Mir et al. 2023](#); [Song, Hwang, and Sheu 2005](#)). Polyvinyl Alcohol fibers are characterized by excellent adhesion to the cement matrix, so they offer great reinforcement against the propagation of micro-cracks ([Ying and Pizhong 2011](#)).

Steel fibers have very good tensile strength and improve mechanical resistance characteristics of cement. In general, fibers are most effective in improving toughness, impact resistance, and dynamic loads resistance of cement ([Mohammad et al. 2022b](#); [Xin, Qinghua, and Shilang 2020](#)). Because glass fibers are strong yet lightweight, and somewhat inert to chemical attack, they represent the most viable candidate to enhance the mechanical performance of cement in such extreme downhole environments ([Babar and Ali 2019](#); [Mazen and Mohammed M.Z 2019](#)). Carbon fibers have a very high tensile strength/stiffness and excellent resistance against chemical and thermal degradation. Besides, they feature light weight and durability ([Khandekar et al. 2019](#); [Shubhangi and Jaiswal 2022](#); [X. Zhang et al. 2007](#)).

The following [table 4](#) shows the overview of the types of fibers used in cement formulations for geothermal application. It discusses various types of fibers, such as polypropylene, nylon, and polyvinyl alcohol (PVA) fibers, which are known for improving crack resistance, toughness, and adhesion properties of cement. Steel fibers have high tensile strength and resistance to impact; glass fibers enhance the performance of cement in chemically aggressive environments. Carbon fibers have exceptionally high tensile strength and resistance to chemicals and thermal loads, therefore suitable under extreme geothermal conditions. The table provides temperature and pressure ranges for each type of fiber, therefore serving to determine their applicability for different geothermal conditions.

Table 4—Synthetic and Metallic Fibers for Enhancing Mechanical Properties of Fiber-Reinforced Cement

Fiber Type	Description	References	Key Benefits	Temperature Range	Pressure Range
Polypropylene Fibers	Polypropylene fibers are lightweight, chemically resistant, and economical. They enhance cement toughness and crack resistance without significantly affecting cement density.	Hua et al. (2018); Jingjun et al. (2016)	Enhances toughness and crack resistance without adding much weight	Up to 180°C (for geothermal applications)	Up to 15,000 psi (for geothermal conditions)
Nylon Fibers	Nylon fibers provide high tensile strength and flexibility, improving toughness, crack resistance, impact resistance, and ductility of cement.	Mir et al. (2023); Song, Hwang, and Sheu (2005)	Increases tensile strength, impact resistance, and ductility	Up to 200°C (for geothermal applications)	Up to 15,000 psi (for geothermal conditions)
Polyvinyl Alcohol (PVA) Fibers	PVA fibers offer excellent adhesion to the cement matrix, providing reinforcement against micro-crack propagation.	Ying and Pizhong (2011)	Reinforces cement against micro-cracks	Up to 250°C (for geothermal applications)	Up to 20,000 psi (for geothermal conditions)

Steel Fibers	Steel fibers have excellent tensile strength and improve mechanical resistance, enhancing toughness and impact resistance, especially under dynamic loads.	Mohammad et al. (2022b); Xin, Qinghua, and Shilang (2020)	Enhances toughness, mechanical resistance, and impact resistance	Up to 300°C (for geothermal applications)	Up to 25,000 psi (for geothermal conditions)
Glass Fibers	Glass fibers are strong, lightweight, and resistant to chemical attack, making them ideal for improving mechanical performance in extreme downhole environments.	Babar and Ali (2019); Mazen and Mohammed M.Z (2019)	Enhances mechanical performance, resistant to chemical attack	Up to 250°C (for geothermal applications)	Up to 15,000 psi (for geothermal conditions)
Carbon Fibers	Carbon fibers have very high tensile strength, stiffness, and excellent resistance to chemical and thermal degradation. They are lightweight and durable, making them ideal for extreme conditions.	Khandekar et al. (2019); Shubhangi and Jaiswal (2022); X. Zhang et al. (2007)	Increases tensile strength, resistance to chemical and thermal degradation	Up to 350°C (for geothermal applications)	Up to 30,000 psi (for geothermal conditions)

Performance Evaluation and Testing

Laboratory Testing of New Cement Formulations

The general discussion of various test required for the cement slurry are discussed below. Note that all the slurry prepared is HPHT conditioned before testing to ensure the slurry testing in reference to the geothermal oil and gas well conditions that can also tolerate HPHT.

Compressive strength test ascertains the capability of the cement in withstanding downhole pressures. For compressive strength test, the cement samples are then under uniaxial compression, and the load at which failure occurs is recorded ([Katherine et al. 2022](#); [L. Yan et al. 2021](#)). Thickening time test ascertains the time available for placing cement before attaining hard setting. During this test under conditions of

downhole pressure and temperature, the consistency of the cement is monitored versus time (Kleef and Vliet 1993; Samuel et al. 2019).

Fluid loss test defines the amount of water lost from the slurry during placement. In this test, Cement slurry undergoes a high-pressure filter press, and then the amount of water lost is measured (Thomas, Daud, and Jun 2019). Tensile strength test determines the resistance of cement to tensile forces. Loads are applied in tension until failure of cements to record the tensile strength (Fahad et al. 2020; Joško et al. 2014; Xudong, Shengxing, and Jikai 2014).

Expansion testing checks the capability of the cement to expand slightly during setting and thus providing good zonal isolation. Volume changes in cement slurries are monitored during hydration (Carpenter 2022). Durability Testing under HPHT Conditions assesses the capability of the cements to retain integrity under HPHT conditions. In this case, cement samples are exposed to high temperature and pressures that would be expected to be experienced under geothermal conditions over an extended duration.

In acid and chemical resistance tests, the reaction of the Cement is measured when it meets acidic or corrosive fluids, including hydrogen sulfide or CO₂. Samples are immersed in corrosive fluids; in this, the change in physical properties is measured for strength and integrity (Barbara G.K et al. 2011; Chunqin et al. 2023).

Permeability Test determines the set cement permeability is interrogated to ensure the sufficiency of sealing. Cement samples are subjected to permeability measurements concerning how much fluid can pass through them (Appleby and Wilson 1996; Lydon and Mahawish 1991; Picandet et al. 2011). Bond Strength Test-shear bond tests the bonding between cement and casing or between cement and formation. By, applying a shear force to the cement and measuring the force that breaks the bond (Abdelfattah et al. 2021; Mohammadreza et al. 2022).

Rheology testing measures the flowing properties of cement slurry. Viscosity testing of the cement slurry under different shear rates to simulate well conditions (Anjuman and Moncef L.N 2011; Chengcheng et al. 2020). Thermal conductivity test assesses the ability of the cement to conduct heat. The specimens are subjected to different temperatures by means of special equipment that simulates well conditions to measure thermal conductivity (Alfonso and Gilberto 2004; Fengyan and Lixin 2022a).

Field Trials and Case Studies

One of the methods that were adopted to overcome the complications brought about by high temperature geothermal wells was foamed cementing at Olkaria geothermal field in Kenya (Stefán et al. 1990). Foamed cement is used in this method, therefore improving thermal insulation characteristics of the cement slurry. By reducing the density of the slurry, the foamed cement offers minimum formation damage and hence is very useful in geothermal environments (Hailin et al. 2023; Wei, Chengwen, and Haoxin 2022). The foamed cement played a crucial role in terms of insulation properties for the wellbore and preventing heat transfer to the surface casing, thereby prolonging the operational life (Tarasov et al. 2010). One of the major lessons learned from this field application was that the foamed cement was highly effective in minimizing the thermal cycling effect often responsible for casing failures in high temperature geothermal wells (Yi et al. 2019). This technique exhibited much potential for extended well longevity at extreme conditions.

Performance Analysis of Foamed Cementing in Olkaria Geothermal Field. Foamed cementing in the Olkaria Geothermal Field revealed several distinct advantages over traditional cementing techniques at least in high-temperature geothermal settings. Among the chief benefits that characterize performance supremacy include:

This may lead to the reduction of thermal conductivity due to the presence of foamed cement in the slurry—a critical problem in geothermal wells. Traditional cement systems, such as Portland cement, tend to conduct heat rather easily; therefore, heat is transferred to the surface casing (Fengyan and Lixin 2022b; Khizar, Romero, and Catalin 2023). This usually causes the casing to expand and then later shrink because

of the imposed stresses, which can create failures over time. In contrast, foamed cement provided quite reasonable thermal insulation against this heat transfer and consequently reduced stress in the casing system (She et al. 2013).

Foamed cement has much lower density compared to conventional cement slurries. This factor is very significant in geothermal wells since weak formations are common (Afshin et al. 2020). In these situations, traditional cementing methods might develop high hydrostatic pressures that could result in either the fracture of formation or losses. In the case of foamed cement, with reduced density, these risks were at a minimum as it maintained the integrity of the well without causing any damage to the formation (Danilo et al. 2019; Nana et al. 2023; Nikoo, Amir, and Ian 2018).

The thermal cycling stress imposed by natural temperature fluctuations in any given geothermal well exerts considerable force on the casing and cement sheath (Honglin et al. 2018; Yuxing, Harshkumar, and Saeed 2021). These constant expansion and contraction cycles often overcome conventional cements because of microcracking, debonding, and fluid migration. In contrast, foamed cement, being inherently elastic and of lower stiffness, absorbed these stresses rather than transferring them to the well structure and causing damage (Harms and Febus 1985).

Foamed cement improved thermal insulation and flexibility, hence improving the well's longevity in the Olkaria field. With conventional techniques, it has typically involved more frequent remedial work driven by casing failures, micro-annuli formation, or degradation of cement at extreme temperatures. Wells which were cemented with the use of the foamed system showed increased resistance to such problems, hence few repairs and less downtime.

Table 5—Comparison of Foamed Cement vs. Traditional Cement (e.g., Portland) in Well Operations

Parameter	Foamed Cement	Traditional Cement (e.g., Portland)
Thermal Insulation	High, reducing heat transfer to surface casing	Low, causing heat migration and stress on casing
Density	Lower, minimizes formation damage and hydrostatic pressure	Higher, risking fractures and losses in weak formations
Thermal Cycling Resistance	Superior, due to elasticity and flexibility	Poor, prone to cracking, debonding, and micro-annuli formation
Well Longevity	Longer operational lifespan with less maintenance	Shorter, due to frequent thermal damage
Cost Efficiency	Higher upfront cost but lower long-term remediation expenses	Lower upfront cost but higher long-term repair costs

Environmental and Safety Considerations

Improvements in cementing techniques have greatly benefited environmental concerns in the operation of geothermal wells, as well as other related subsurface operations. These techniques afford better well integrity and performance with protection of the environment from noxious emissions and minimizing risks to groundwater contamination. Some of the key environmental benefits include:

Reduced Fluid Migration

Groundwater Contamination Prevention: Improvement in cementing practices, such as the development of self-healing cements and expanding agents, offers a much superior seal between the wellbore and formations. This reduces the possibility of fluid migration through the annular space, which can prevent contaminants from leaking into freshwater aquifers. Protection of groundwater is an important environmental issue, especially in geothermal and oil and gas operations (Jackson et al. 2013; Putra et al. 2021).

Improved Zonal Isolation. Better cement placement and the introduction of advanced materials contribute a great deal toward better zonal isolation, which, among other things, aims at preventing the flow of unwanted formation fluids or gas across different geological layers. This means the geothermal fluids or hydrocarbons remain confined to their target zones to minimize potential environmental impacts associated with fluid leakage (Adnan et al. 2023; Foroushan et al. 2021).

Reduced CO₂ Emissions

Less CO₂ leakage. As cementing improves, so does long-term containment in injected CO₂-especially in CO₂ sequestration wells. Quality cement sheaths impede pathways of leaks that could release the gas into the atmosphere, improving the effectiveness of CCS projects for mitigating climate change (Barlet-Gouédard et al. 2007; Catalin and Opeyemi 2020).

Blended Cements with Lower Carbon Footprint. Blended Portland cements utilize supplementary cementing materials such as fly ash, slag, and pozzolans to reduce demand for pure Portland cement production, a very energy-intensive process. These alternative materials, therefore, lower the overall carbon footprint of the cement to be used in geothermal wells, hence helping in the reduction of CO₂ emissions from the cement industry (Gartner 2004; Tatiana, Víctor, and Julián 2014).

Optimal Cementing Practices. Better cementing techniques reduce the amount of remedial cementing jobs. Lesser numbers of re-drills and repairs mean less energy use and lower CO₂ emission related to drilling and cementing activities.

Extended Well Life

Minimization of Re-drilling and Workovers

The improved resistance and performance of the cement sheaths reduce well repairs and re-drillings to less frequently. This, in turn, prolongs the service life of geothermal wells due to the consumption of fewer resources and less energy for maintenance, hence minimizing the environmental burden over time (Luswata, Onyancha, and Thuo 2023).

Reduced Resource Extraction. By not having to replace as many wells, the requirement for raw materials like steel and cement and other construction materials is lower, thus reducing the footprint of the well over its lifecycle.

Improved Thermal Efficiency

Enhanced Insulation Properties: Improved thermal insulation properties with foamed cement and other new materials reduce the loss of heat to the surrounding formations. This increases geothermal well efficiency, where more energy can be captured with each unit of input; hence, it reduces overall energy demand and resultant emissions (Sheng et al. 2020b).

Reduced Emissions of Hazardous Elements

Minimized Methane and Hazardous Gas Leakage. Quality cementing decreases the possibility of leaking methane or any other hazardous gases into the atmosphere. Methane is one of the strongest greenhouse gases, and the accidental leakage during geothermal or hydrocarbon extraction might result in serious environmental consequences. Improved cement integrity can also avoid that kind of leak and will protect the atmosphere from harmful emissions (Abdelmjeed et al. 2022; Nowamooz et al. 2015).

Carbon Capture and Storage (CCS)

Sequestration Efficiency. Advanced cementing technologies link wellbore structural integrity to the long-term needs of CO₂ sequestration. The cement systems will ensure that structural support is maintained to withstand the chemical and thermal stresses driven by CO₂ storage, thus ensuring securely stored carbon underground and adding to the worldwide atmospheric reduction of CO₂ levels (Qingyun et al. 2015).

Reduced Energy Consumption in Cement Production

Alternative Binders and Additives Usage. In concreting applications, the usage of alternative binders with low-energy additives like SCMs reduces the energy used to produce cements. Subsequently, this reduces the carbon footprint created by the cementing materials applied in service, as a contribution to a global goal of decarbonization of the energy and industrial sectors (Golewski 2021).

Minimization of Surface and Subsurface Disturbances. **Fewer Surface Disturbances:** Advanced cementing reduces the chances of failures; therefore, fewer surface interventions regarding re-drilling or well workovers. This diminishes the need to mobilize extra equipment and, as such, reduces surface disturbance and possible environmental impact to ecosystems and landscapes.

Future Areas of Study and Guidance on Geothermal Well Cementing

Fields for Further Research and Development

Formulation Development of Cement Compositions for Geothermal Wells. *Investigation of New Cement Chemistries and Materials:* Research into innovative cement formulations, including the use of nanomaterials and geopolymer cements, could significantly improve the thermal and chemical resistance of cement under high-temperature and high-pressure (HTHP) conditions typical of geothermal reservoirs. Studies should focus on enhancing the durability of cement formulations, making them more compatible with varying geothermal reservoir conditions, and improving resistance to both high temperatures (above 300°C) and corrosive fluids such as CO₂.

Self-Healing Cements: Develop self-healing cement formulations for geothermal wells that can self-heal microcracks automatically caused by temperature variation, pressure variation, and mechanical stress. This will enhance the service life and integrity of the cement sheaths in the geothermal environment and ensure that there is no leak or failure.

Long-Term Performance Assessment in Geothermal Conditions. Long-term field tests for the cement behavior: There are long-term field trials meant to test the performance of the geothermal well cements under extreme temperature and pressure exposure with fluids. Great emphasis is to be given to how much resistance cement has towards the geothermal fluids including the brines and acids as well as its ability in the long duration, say decades of geothermal well operations.

Monitoring Thermal and Mechanical Stress: Determine how the thermal time-dependent fluctuations in geothermal wells influence cement performance under long-term exposure in high pressure reservoirs, and how cement systems behave under repeated thermal cycling

Modeling and Simulation of Geothermal Wells. Advanced Computational Models: Develop advanced models that simulate the behavior of cement under geothermal well conditions, considering thermal expansion, shrinkage, chemical degradation, and mechanical stress. Such models should be able to account for real-world dynamic geothermal environment, such as pressure and temperature fluctuations, as well as fluid composition changes, to predict possible failure modes and aid in designing the optimal cement slurries.

Potential for Digital Technologies in Geothermal Cementing.

AI and Machine Learning for Geothermal Cement Design:

AI for Geothermal Cement Slurry Optimization: Use AI-based models to design geothermal well optimized cement slurries. This provides an understanding of the cement behavior under conditions of the geothermal reservoir, considering various parameters of temperature, pressure, and fluid chemistry for a much more efficient operation of the cementing.

Machine Learning for Cement Failure Prediction: Use machine learning algorithms on geothermal well data to find some patterns that can predict when cement failure will occur. Historic data, like well temperature, pressure cycles, and the composition of cement, could be scanned to predict and prevent potential problems.

Real-Time Monitoring and Diagnostics for Geothermal Wells:

IoT-enabled Sensors for Cement Integrity: Install sensors with IoT in geothermal wells, providing continuous integrity and bond strength of cement in the wells. In future, the sensors will monitor at high temperature and pressure at which cement sheath degrades and alerts the operator of the presence of anomalies such as leaks, fracture, or weakening of cement matrix.

Digital Twins of Geothermal Wells: Develop digital twins of geothermal well structures to simulate cement performance under real-time conditions. The simulation may provide insight into how the cement performs under operational pressures, temperature fluctuations, and fluid exposure.

Automation in Cementing Operations for Geothermal Wells:

Robotics and Automation for Cement Placement: Apply robotics and automation technologies to enhance accuracy of cement placement in geothermal wells. This will likely be a critical pathway for achieving consistent cementing in more challenging geothermal reservoirs and to limit human errors, optimizing efficiency in cement jobs.

Recommendations for Industry Adoption in Geothermal Well Cementing.

Joint Research Projects for Geothermal Cementing:

Cross-Sector Cooperation: Academia, geothermal industry experts, and government agencies should collaborate to enhance development and adoption of advanced cementing technologies specifically designed for geothermal wells. Mutual research on new materials, technologies, and operational strategies can be facilitated through this cross-sector cooperation.

Shared Databases for Geothermal Cementing Performance:

Establish a centralized database that summarizes performance information, failure case histories, and research on geothermal well cementing. In sharing it across the industry, it will enable informed decisions and will be indispensable in optimizing the cementing practice.

Standardization and Guidelines for Geothermal Cementing.

Update Regulatory Frameworks:

Encourage regulatory bodies to update standards to include advanced cement formulations that are resistant to geothermal conditions, including high temperatures, pressures, and corrosive fluids. Guidelines should also encompass the use of digital technologies for monitoring and controlling cement performance in real time.

Adoption of International Standards: Promote the use of international standards in geothermal cementing operations to ensure consistency, quality, and safety. Such standards should be updated as there are improvements in cementing technologies and operational practices in geothermal reservoirs.

Training and Capacity Building for Geothermal Cementing:

Advanced Training for Geothermal Well Cementing on AI and Digital Monitoring: Train personnel involved in geothermal well cementing on the usage of AI-driven tools, real-time monitoring systems, and digital twins. This will make them aware of and take full advantage of new technologies for improving cement placement and maintaining well integrity.

Continuous Education Programs: Develop continuous education and professional development programs aimed at the latest advances in geothermal cementing: self-healing cements, nanomaterials, and digital technologies.

Sustainability Considerations for Geothermal Well Cementing:

Green Cements for Geothermal Wells: Develop low carbon intensity cements and more sustainable cementing practices that reduce the adverse environmental footprint of geothermal well construction. Alternative material research into geopolymers or pozzolanic cements could significantly impact the reduction of carbon emission associated with cement production.

Lifecycle analysis of geothermal well cements: Perform lifecycle analyses to assess the environmental implications and sustainability of geothermal cementing materials in wells. Such analysis should consider the complete lifecycle of geothermal wells from drilling to abandonment stages. Cementing practices during all these stages should then align with sustainability goals.

Lessons Learned in Geothermal Well Cementing

Key Takeaways

Geothermal reservoirs: Geothermal wells are exposed to extreme temperatures, from 350°C, and chemically aggressive environments, and therefore specialized formulations of cement are necessary to endure such conditions. Advancements in cement technology, self-healing cements, nanomaterial additives (such as nano-silica, carbon nanotubes, and graphene oxide), and AI-driven design have proven effective in overcoming the hurdles of microcracking, thermal stress, and fluid migration.

Real-Time Monitoring is Critical for Well Integrity: The use of IoT-enabled sensors and digital twins for continuous real-time monitoring of well cement integrity has become a critical strategy to ensure long-term stability of geothermal wells. It helps in the early identification of potential problems and aids predictive maintenance, which increases the life of geothermal wells.

Importance of Innovative Cementing Techniques for Geothermal Wells

Improvement in Geothermal Well Reliability: Innovative Cementing Techniques In the enhancement of geothermal well reliability, innovations in cementing techniques have taken center stage. These innovations improve the mechanical properties and thermal stability and chemical resistance of these wells, to a greater extent, decrease the failure risk of a well, and reduce the cost associated with maintenance for continued production of geothermal energy.

New prospects for optimizing geothermal well cementing have been brought by digital technologies, especially AI and IoT. AI supports the design of cement slurries optimized for specific conditions in geothermal wells. IoT-based monitoring provides essential real-time data, thereby enabling proactive decision-making. These digital innovations elevate the operational excellence and performance optimization in geothermal projects.

Impact on the Future of Geothermal Energy Production

Scaling and Sustaining Geothermal Energy: Advanced cementing technologies will help scale and then sustain the production of geothermal energy. New advancements in cement will mean well integrity and long-term durability and minimize the actual downtime from operational reasons so that it can provide positive economics to the projects geothermal in nature. Those innovations would position geothermal energy as a more competitive and reliable source for renewable power in the global market.

Environmental Benefits: The adoption of these advanced cementing technologies also aligns with the geothermal sector's broader sustainability objectives. By ensuring well integrity and reducing environmental impact, these technologies help the industry progress toward carbon neutrality.

The integration of these cementing solutions is crucial to realizing the full potential of geothermal energy. As geothermal wells become more reliable and efficient, these innovations will make way for a greener, more sustainable future in energy production.

In conclusion, the lessons learned from these advancements in geothermal well cementing underscore the potential of these technologies to bring about transformation in improving well integrity, optimizing geothermal production, and contributing toward a sustainable energy future.

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